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# basic education

Department:  
Basic Education  
**REPUBLIC OF SOUTH AFRICA**

**SENIOR CERTIFICATE EXAMINATIONS/  
SENIORSERTIFIKAAT-EKSAMEN  
NATIONAL SENIOR CERTIFICATE EXAMINATIONS/  
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

**MATHEMATICS P1/WISKUNDE V1**

**MARKING GUIDELINES/NASIENRIGLYNE**

**2022**

**MARKS: 150  
PUNTE: 150**

**These marking guidelines consist of 16 pages.  
*Hierdie nasienriglyne bestaan uit 16 bladsye.***

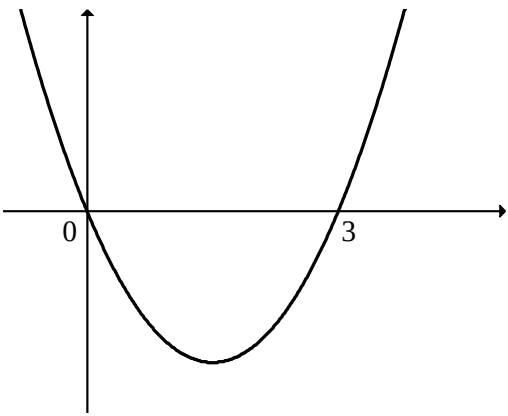
**NOTE:**

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- Consistent Accuracy applies in all aspects of the marking guidelines.

**LET WEL:**

- Indien 'n kandidaat 'n vraag TWEE keer beantwoord, merk slegs die EERSTE poging.
- Volgehoue akkuraatheid is DEURGAANS op ALLE aspekte van die nasienriglyne van toepassing.

**QUESTION/VRAAG 1**

|       |                                                                                                                                                                                       |                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1.1.1 | $x^2 + 2x - 15 = 0$<br>$(x + 5)(x - 3) = 0$<br>$x = -5$ or $x = 3$                                                                                                                    | ✓ factors<br>✓ $x = -5$<br>✓ $x = 3$<br>(3)                                         |
| 1.1.2 | $5x^2 - x - 9 = 0$<br>$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(5)(-9)}}{2(5)}$<br>$x = \frac{1 \pm \sqrt{181}}{10}$<br>$x = 1,45$ or $x = -1,25$                                         | ✓ substitution into the correct formula<br><br>✓ $x = 1,45$<br>✓ $x = -1,25$<br>(3) |
| 1.1.3 | $x^2 \leq 3x$<br>$x^2 - 3x \leq 0$<br>$x(x - 3) \leq 0$<br><br>$0 \leq x \leq 3$ OR $x \in [0; 3]$ | ✓ standard form<br>✓ factors<br><br><br><br><br>✓✓ answer<br>(4)                    |
| 1.2.1 | $a + \frac{64}{a} = 16$<br>$a^2 - 16a + 64 = 0$<br>$(a - 8)^2 = 0$<br>$a = 8$                                                                                                         | ✓ standard form<br>✓ factors<br><br>✓ answer<br>(3)                                 |



**OR/OF**

$$x = \frac{2+y}{2} \quad \dots(1)$$

$$\frac{1}{x} - 3y = 1 \quad \dots(2)$$

$$\frac{\frac{1}{\frac{2+y}{2}} - 3y = 1$$

$$\frac{2}{2+y} - 3y = 1$$

$$\frac{2 - 6y - 3y^2}{2+y} = 1$$

$$2 - 6y - 3y^2 = 2 + y$$

$$-3y^2 - 7y = 0$$

$$-y(3y + 7) = 0$$

$$y = 0 \quad \text{or} \quad y = -\frac{7}{3}$$

$$x = 1 \quad \text{or} \quad x = -\frac{1}{6}$$

**OR/OF**

$$\checkmark x = \frac{2+y}{2}$$

✓ substitution

✓ simplification

✓ standard form

✓ y-values

✓ x-values

(6)  
[26]

## QUESTION/VRAAG 2

|       |                                                                                                                                                                                                                                                 |                                                                                                              |                                                                                                  |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 2.1.1 | $a + 6d = 35$<br>$-1 + 6d = 35$<br>$6d = 36$<br>$d = 6$<br><b>OR/OF</b><br>$\frac{35 - (-1)}{7 - 1} = 6$                                                                                                                                        | <div style="border: 1px solid black; padding: 5px; display: inline-block;">ANSWER ONLY:<br/>FULL MARKS</div> | ✓ substitution<br><br>✓ answer (2)<br><b>OR/OF</b><br>✓ substitution<br>✓ answer (2)             |
| 2.1.2 | $T_n = a + (n - 1)d$<br>$473 = -1 + (n - 1)(6)$<br>$79 = n - 1$<br>$\therefore n = 80$                                                                                                                                                          | <div style="border: 1px solid black; padding: 5px; display: inline-block;">ANSWER ONLY:<br/>FULL MARKS</div> | ✓ substitution into the correct formula<br>✓ equating to 473<br>✓ answer (3)                     |
| 2.1.3 | $S_n = \frac{n}{2}[2a + (n - 1)d]$<br>$S_{40} = \frac{40}{2}[2(-1) + (40 - 1)(6)]$<br>$\therefore S_{40} = 4640$<br><br><b>OR/OF</b><br>$T_{40} = 6(40) - 7$<br>$= 233$<br>$S_n = \frac{n}{2}(a + l)$<br>$= \frac{40}{2}(-1 + 233)$<br>$= 4640$ |                                                                                                              | ✓ substitution<br><br>✓ answer (2)<br><br><b>OR/OF</b><br><br><br>✓ substitution<br>✓ answer (2) |
| 2.2.1 | <br>$T_5 = 11$                                                                                                                                                                                                                                  |                                                                                                              | ✓ answer (A) (1)                                                                                 |
| 2.2.2 | $T_n = an^2 + bn + c$<br>$2a = 4$<br>$a = 2$<br>$3a + b = -22$<br>$6 + b = -22$<br>$b = -28$<br>$a + b + c = 75$<br>$2 - 28 + c = 75$<br>$c = 101$<br>$\therefore T_n = 2n^2 - 28n + 101$                                                       |                                                                                                              | ✓ $T_n = an^2 + bn + c$<br><br>✓ $a = 2$<br>✓ $b = -28$<br><br>✓ $c = 101$ (4)                   |

2.2.3 Minimum value of  $T_n$ 

$$n = -\frac{b}{2a} = -\frac{(-28)}{2(2)}$$

$$n = 7$$

$$\checkmark n = 7$$

$$\text{Minimum value of } T_n = 2(7)^2 - 28(7) + 101 = 3$$

$$\checkmark \text{ min value} = 3$$

Each term in the new pattern is  $-\frac{1}{5}$  the value of the terms in the old pattern.

$$\checkmark -\frac{1}{5} \text{ value of term of old pattern}$$

$$\text{Maximum value of new pattern} = -\frac{3}{5}$$

$$\checkmark \text{ max value} = -\frac{3}{5} \quad (4)$$

**OR/OF**

$$T'_n = 4n - 28$$

$$4n - 28 = 0$$

$$4n = 28$$

$$n = 7$$

**OR/OF**

$$\checkmark n = 7$$

$$\text{Minimum value of } T_n = 2(7)^2 - 28(7) + 101 = 3$$

$$\checkmark \text{ min value} = 3$$

Each term in the new pattern is  $-\frac{1}{5}$  the value of the terms in the old pattern.

$$\checkmark -\frac{1}{5} \text{ value of term of old pattern}$$

$$\text{Maximum value of new pattern} = -\frac{3}{5}$$

$$\checkmark \text{ max value} = -\frac{3}{5} \quad (4)$$

**OR/OF**

$$T_n = -\frac{2}{5}n^2 + \frac{28}{5}n - \frac{101}{5}$$

**OR/OF**

$$\checkmark \checkmark T_n \div (-5)$$

$$n = -\frac{b}{2a} = \frac{-\frac{28}{5}}{2\left(\frac{-2}{5}\right)}$$

$$= 7$$

$$\checkmark n = 7$$

$$T_7 = -\frac{3}{5}$$

$$\checkmark \text{ max value} = -\frac{3}{5} \quad (4)$$

**OR/OF****OR/OF**

$$T_n = -\frac{2}{5}n^2 + \frac{28}{5}n - \frac{101}{5}$$

$$T'_n = -\frac{4}{5}n + \frac{28}{5}$$

$$\checkmark\checkmark T_n \div (-5)$$

$$-\frac{4}{5}n + \frac{28}{5} = 0$$

$$-4n = -28$$

$$n = 7$$

$$\text{Minimum value of } T_n = 2(7)^2 - 28(7) + 101 = 3$$

Each term in the new pattern is  $-\frac{1}{5}$  the value of the terms in the old pattern.

$$\checkmark n = 7$$

$$\text{Maximum value of new pattern} = -\frac{3}{5}$$

$$\checkmark \text{ max value} = -\frac{3}{5}$$

(4)

[16]



**QUESTION/VRAAG 3**

|       |                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                             |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.1.1 | $T_n = ar^{n-1}$<br>$T_{10} = 1\,024 \left(\frac{1}{4}\right)^{10-1}$<br>$\therefore T_{10} = \frac{1}{256}$ <div style="border: 1px solid black; padding: 5px; display: inline-block; margin-left: 20px;"> <b>ANSWER ONLY:<br/>FULL MARKS</b> </div>                                                                                                                                                            | ✓ substitution into the correct formula<br>✓ answer<br>(2)                                                                                                                                                  |
| 3.1.2 | $\sum_{p=0}^8 256(4^{1-p}) = 1\,024 + 256 + 64 + \dots$<br>$S_n = \frac{a[1-r^n]}{1-r}$<br>$S_9 = \frac{1\,024 \left[1 - \left(\frac{1}{4}\right)^9\right]}{1 - \frac{1}{4}}$<br>$S_9 = \frac{87\,381}{64}$<br>$= 1\,365,33$<br><br><b>OR/OF</b><br>$\sum_{p=0}^8 256(4^{1-p})$<br>$= 1\,024 + 256 + 64 + 16 + 4 + 1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64}$<br>$S_9 = \frac{87\,381}{64}$<br>$= 1\,365,33$ | ✓ 1024<br><br>✓ $n = 9$<br>✓ substitution into the correct formula<br><br>✓ answer<br>(4)<br><br><b>OR/OF</b><br><br>✓ 1024<br>✓ rest of expansion<br>✓ $n = 9$ terms<br>✓ answer<br>(4)                    |
| 3.2   | $-t^2 - 6t - 9; \frac{t^3 + 9t^2 + 27t + 27}{2}$<br>$-(t^2 + 6t + 9); \frac{1}{2}(t+3)(t^2 + 6t + 9)$<br>$-(t+3)^2; \frac{1}{2}(t+3)^3$<br>$r = \frac{-(t+3)}{2}$<br>$-1 < \frac{-t-3}{2} < 1$<br>$-2 < -t-3 < 2$<br>$1 < -t < 5$<br>$-5 < t < -1$                                                                                                                                                               | $\frac{t^3 + 9t^2 + 27t + 27}{-t^2 - 6t - 9}$<br>✓ $r = \frac{2}{-t^2 - 6t - 9}$<br><br>✓ $-(t^2 + 6t + 9)$<br>✓ $\frac{1}{2}(t+3)(t^2 + 6t + 9)$<br><br>✓ $-1 < \frac{-t-3}{2} < 1$<br><br>✓ answer<br>(5) |
|       |                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>[11]</b>                                                                                                                                                                                                 |

**QUESTION 4**

|       |                                                                                                                                                                                                                                                                                 |                                                        |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 4.1   | $10 = a\left(\frac{1}{3}\right)^{-2} + 7$ $3 = 9a$ $\therefore a = \frac{1}{3}$                                                                                                                                                                                                 | ✓ subs $(-2 ; 10)$<br>✓ simplification<br>✓ answer (3) |
| 4.2   | $y = g(0)$ $y = \frac{1}{3} \times \left(\frac{1}{3}\right)^0 + 7$ $y = \frac{22}{3} = 7,33$ $\therefore \left(0 ; \frac{22}{3}\right)$ <div style="border: 1px solid black; padding: 5px; display: inline-block; margin-top: 10px;">ANSWER ONLY:<br/>FULL MARKS</div>          | ✓ substitution of $x = 0$<br>✓ answer (2)              |
| 4.3.1 | Translation by 1 unit to the right and 7 units downwards                                                                                                                                                                                                                        | ✓ 1 unit right<br>✓ 7 units downwards (2)              |
| 4.3.2 | $h(x) = \left(\frac{1}{3}\right)^x$ $h^{-1}: x = \left(\frac{1}{3}\right)^y$ $y = \log_{\frac{1}{3}}(x) \quad \text{OR/OF} \quad y = -\log_3(x)$ <div style="border: 1px solid black; padding: 5px; display: inline-block; margin-top: 10px;">ANSWER ONLY:<br/>FULL MARKS</div> | ✓ swap $x$ and $y$<br>✓ answer (2)                     |
|       |                                                                                                                                                                                                                                                                                 | <b>[9]</b>                                             |

## QUESTION 5

|     |                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                 |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.1 | $g(x) = \frac{a}{x+2} + q$<br>Subs (1 ; 0):<br>$0 = \frac{a}{1+2} + q$<br>$0 = a + 3q$<br>Subs $\left(0 ; -\frac{1}{2}\right)$<br>$-\frac{1}{2} = \frac{a}{0+2} + q$<br>$-1 = a + 2q$<br>Solving simultaneously:<br>$q = 1$<br>$a = -3$<br>$\therefore g(x) = \frac{-3}{x+2} + 1$ | $\checkmark g(x) = \frac{a}{x+2} + q$<br><br>$\checkmark 0 = a + 3q$<br><br>$\checkmark -1 = a + 2q$<br>$\checkmark$ solving simultaneously<br>$\checkmark q = 1$<br>$\checkmark a = -3$<br>(6) |
| 5.2 | $y \in \mathbb{R}; y \neq 1$<br><b>OR/OF</b><br>$(-\infty; 1)$ or $(1; \infty)$<br><b>OR/OF</b><br>$y < 1$ or $y > 1$                                                                                                                                                             | $\checkmark$ answer<br>(1)                                                                                                                                                                      |
| 5.3 | $y - 1 = 1(x + 2)$ <b>OR/OF</b> $1 = 1(-2) + c$<br><div style="border: 1px solid black; padding: 5px; display: inline-block; margin: 10px;"> ANSWER ONLY:<br/> FULL MARKS </div> $c = 3$<br>$y = x + 3$                                                                           | $\checkmark m = 1$<br>$\checkmark$ subs point $(-2; 1)$<br>$\checkmark$ answer<br>(3)                                                                                                           |
| 5.4 | $K'(-3; 4)$                                                                                                                                                                                                                                                                       | $\checkmark$ x-value<br>$\checkmark$ y-value<br>(2)                                                                                                                                             |
|     |                                                                                                                                                                                                                                                                                   | <b>[12]</b>                                                                                                                                                                                     |



**QUESTION 7**

|       |                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                         |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.1   | $A = P(1+i)^n$<br>$2 = 1\left(1 + \frac{0,085}{4}\right)^{4n}$<br>$4n = \log_{\left(1 + \frac{0,085}{4}\right)} 2$<br>$n = 8,24 \text{ years}$                                                                            | $\left. \begin{array}{l} \checkmark 2 \\ \checkmark \frac{0,085}{4} \end{array} \right\} \begin{array}{l} \text{In} \\ \text{correct} \\ \text{formula} \end{array}$<br>$\checkmark \text{ use of logs}$<br>$\checkmark \text{ answer in years}$<br>(4) |
| 7.2.1 | $A = P(1-i)^n$<br>$180\,000 = 500\,000(1-i)^5$<br>$\frac{9}{25} = (1-i)^5$<br>$\sqrt[5]{\frac{9}{25}} = 1-i$<br>$i = 0,1848068\dots$<br>$r = 18,48\%$                                                                     | $\checkmark \text{ subs into correct formula}$<br>$\checkmark \text{ simplification}$<br>$\checkmark i = 0,1848\dots$<br>$\checkmark \text{ answer}$<br>(4)                                                                                             |
| 7.2.2 | $A = P(1+i)^n$<br>$A = 500\,000(1+0,063)^5$<br>$A = R678\,635,11$                                                                                                                                                         | $\checkmark \text{ subs into correct formula}$<br>$\checkmark \text{ answer}$<br>(2)                                                                                                                                                                    |
| 7.2.3 | Sinking Fund = $678\,635,11 - 180\,000$<br>= R 498 635,11<br>$498\,635,11 = \frac{x \left[ \left(1 + \frac{0,1025}{12}\right)^{58} - 1 \right] \left(1 + \frac{0,1025}{12}\right)^3}{\frac{0,1025}{12}}$ $x = R6\,510,36$ | $\checkmark \text{ value of sinking fund}$<br>$\checkmark \frac{0,1025}{12}$<br>$\checkmark n = 58 \text{ (A)}$<br>$\checkmark \left(1 + \frac{0,1025}{12}\right)^3$<br>$\checkmark \text{ answer (A)}$<br>(5)                                          |
|       |                                                                                                                                                                                                                           | <b>[15]</b>                                                                                                                                                                                                                                             |

**QUESTION/VRAAG 8**

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8.1   | $f(x) = -x^2$ $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $f'(x) = \lim_{h \rightarrow 0} \frac{-(x+h)^2 + x^2}{h}$ $f'(x) = \lim_{h \rightarrow 0} \frac{-x^2 - 2xh - h^2 + x^2}{h}$ $= \lim_{h \rightarrow 0} \frac{-2xh - h^2}{h}$ $= \lim_{h \rightarrow 0} \frac{h(-2x - h)}{h}$ $= \lim_{h \rightarrow 0} (-2x - h)$ $\therefore f'(x) = -2x$ <p><b>OR/OF</b></p> $f(x) = -x^2$ $f(x+h) = -(x+h)^2 = -x^2 - 2xh - h^2$ $f(x+h) - f(x) = -x^2 - 2xh - h^2 - (-x^2) = -2xh - h^2$ $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $= \lim_{h \rightarrow 0} \frac{-2xh - h^2}{h}$ $= \lim_{h \rightarrow 0} \frac{h(-2x - h)}{h}$ $= \lim_{h \rightarrow 0} (-2x - h)$ $\therefore f'(x) = -2x$ | <p>✓ substitution into formula</p> <p>✓ <math>-(x^2 + 2xh + h^2)</math></p> <p>✓ <math>-2xh - h^2</math></p> <p>✓ <math>-2x - h</math></p> <p>✓ answer (5)</p> <p><b>OR/OF</b></p> <p>✓ <math>-x^2 - 2xh - h^2</math></p> <p>✓ <math>-2xh - h^2</math></p> <p>✓ substitution into the formula</p> <p>✓ <math>-2x - h</math></p> <p>✓ answer (5)</p> |
| 8.2.1 | $f(x) = 4x^3 - 5x^2$ $f'(x) = 12x^2 - 10x$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>✓ <math>12x^2</math> (A)</p> <p>✓ <math>-10x</math> (A)</p> <p>(2)</p>                                                                                                                                                                                                                                                                           |
| 8.2.2 | $D_x \left[ \frac{-6\sqrt[3]{x} + 2}{x^4} \right]$ $= D_x \left[ \frac{-6(x)^{\frac{1}{3}}}{x^4} + \frac{2}{x^4} \right]$ $= D_x \left[ -6x^{-\frac{11}{3}} + 2x^{-4} \right]$ $= 22x^{-\frac{14}{3}} - 8x^{-5}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>✓ <math>x^{\frac{1}{3}}</math></p> <p>✓ <math>-6x^{-\frac{11}{3}} + 2x^{-4}</math></p> <p>✓ <math>22x^{-\frac{14}{3}}</math></p> <p>✓ <math>-8x^{-5}</math></p> <p>(4)</p>                                                                                                                                                                       |
|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>[11]</b>                                                                                                                                                                                                                                                                                                                                         |

## QUESTION/VRAAG 9

|       |                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9.1   | $f(x) = (x+t)^2(x-3)$<br>$-3 = (0+t)^2(0-3)$<br>$1 = t^2$<br>$t = \pm 1$<br>$\therefore t = 1$<br>$f(x) = (x+1)^2(x-3)$<br>$f(x) = (x^2 + 2x + 1)(x-3)$<br>$f(x) = x^3 - x^2 - 5x - 3$                                                                                                                                                              | $\checkmark f(x) = (x+t)^2(x-3)$<br>$\checkmark$ subs $(0; -3)$<br><br>$\checkmark t$<br><br>$\checkmark f(x) = (x+1)^2(x-3)$<br>$\checkmark$ expansion<br><div style="text-align: right;">(5)</div>                                                                                                      |
| 9.2   | $f'(x) = 3x^2 - 2x - 5$<br>$0 = 3x^2 - 2x - 5$<br>$0 = (x+1)(3x-5)$<br>$x = -1$ or $x = \frac{5}{3}$<br>$N\left(\frac{5}{3}; -\frac{256}{27}\right) = (1,67; -9,48)$                                                                                                                                                                                | $\checkmark f'(x) = 3x^2 - 2x - 5$<br>$\checkmark = 0$<br><br>$\checkmark$ factors<br>$\checkmark$ x-value $(x > 0)$<br><br>$\checkmark$ y-value (A) <span style="float: right;">(5)</span>                                                                                                               |
| 9.3.1 | $x < 3$ ; $x \neq -1$<br><br><b>OR/OF</b><br>$x < -1$ or $-1 < x < 3$<br><br><b>OR/OF</b><br>$(-\infty; -1)$ or $(-1; 3)$                                                                                                                                                                                                                           | $\checkmark x < 3$<br>$\checkmark x \neq -1$ <span style="float: right;">(2)</span><br><b>OR/OF</b><br>$\checkmark x < -1$<br>$\checkmark -1 < x < 3$ <span style="float: right;">(2)</span><br><b>OR/OF</b><br>$\checkmark (-\infty; -1)$<br>$\checkmark (-1; 3)$ <span style="float: right;">(2)</span> |
| 9.3.2 | $x < -1$ or $x > \frac{5}{3}$ OR/OF $x \leq -1$ or $x \geq \frac{5}{3}$<br><br><b>OR/OF</b><br>$(-\infty; -1)$ or $\left(\frac{5}{3}; \infty\right)$ OR/OF $(-\infty; -1]$ or $\left[\frac{5}{3}; \infty\right)$                                                                                                                                    | $\checkmark x < -1$<br>$\checkmark x > \frac{5}{3}$ <span style="float: right;">(2)</span><br><b>OR/OF</b><br>$\checkmark (-\infty; -1)$<br>$\checkmark \left(\frac{5}{3}; \infty\right)$ <span style="float: right;">(2)</span>                                                                          |
| 9.3.3 | $f''(x) > 0$<br>$6x - 2 > 0$<br>$x > \frac{1}{3}$ or $\left(\frac{1}{3}; \infty\right)$<br><br><b>OR/OF</b><br>$\frac{\frac{5}{3} + (-1)}{2} = \frac{1}{3}$<br>$x > \frac{1}{3}$ or $\left(\frac{1}{3}; \infty\right)$ <div style="border: 1px solid black; padding: 5px; width: fit-content; margin: 10px auto;">ANSWER ONLY:<br/>FULL MARKS</div> | $\checkmark 6x - 2$<br>$\checkmark \frac{1}{3}$<br>$\checkmark x > \frac{1}{3}$ <span style="float: right;">(3)</span><br><b>OR/OF</b><br>$\checkmark$ substitution<br>$\checkmark \frac{1}{3}$<br>$\checkmark x > \frac{1}{3}$ <span style="float: right;">(3)</span>                                    |

|     |                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                       |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9.4 | $\text{Distance} = x^3 - x^2 - 5x - 3 - (3x^2 - 2x - 5)$ $= x^3 - 4x^2 - 3x + 2$ $\frac{d\text{Distance}}{dx} = 3x^2 - 8x - 3$ $0 = 3x^2 - 8x - 3$ $0 = (3x + 1)(x - 3)$ $x = 3 \text{ or } x = -\frac{1}{3}$ <p>Max distance</p> $= \left(-\frac{1}{3}\right)^3 - 4\left(-\frac{1}{3}\right)^2 - 3\left(-\frac{1}{3}\right) + 2$ $= \frac{68}{27} = 2,52$ | $\checkmark x^3 - 4x^2 - 3x + 2$<br>$\checkmark \frac{d\text{Distance}}{dx} = 3x^2 - 8x - 3$<br>$\checkmark$ factors<br>$\checkmark$ x-values<br>$\checkmark x = -\frac{1}{3}$<br>$\checkmark$ answer |
|     |                                                                                                                                                                                                                                                                                                                                                            | (6)                                                                                                                                                                                                   |
|     |                                                                                                                                                                                                                                                                                                                                                            | [23]                                                                                                                                                                                                  |



**QUESTION/VRAAG 10**

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                        |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10.1.1 | $7! = 5\,040$                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ✓✓ answer<br>(2)                                                                                                                                                                       |
| 10.1.2 | $4! \times 4!$<br>$= 576$<br>$P(\text{African flags together}) = \frac{576}{5040} \left( = \frac{4}{35} = 0,11 \right)$                                                                                                                                                                                                                                                                                                                                                  | ✓ $4!$<br>✓ $4! \times 4!$<br>✓ answer (A)<br>(3)                                                                                                                                      |
| 10.2   | $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$<br>$0,88 = 0,4 + P(B) - P(A \text{ and } B)$<br>$0,88 = 0,4 + P(B) - 0,4P(B)$<br>$0,48 = 0,6P(B)$<br>$P(B) = 0,8$                                                                                                                                                                                                                                                                                               | ✓ subs into rule<br>✓ $P(A \text{ and } B) = 0,4P(B)$<br>✓ answer<br>(3)                                                                                                               |
| 10.3   | <p>First Passenger      Second Passenger</p> <p>Probability of first passenger choosing meat <math>= \frac{x}{120}</math></p> <p>Probability of second passenger choosing cheese <math>= \frac{120-x}{119}</math></p> $\frac{x}{120} \times \frac{120-x}{119} = \frac{18}{85}$ $120x - x^2 = 3\,024$ $x^2 - 120x + 3\,024 = 0$ $(x - 84)(x - 36) = 0$ $x = 84 \quad \text{or} \quad x = 36$ $\therefore P(1^{\text{st}} \text{ cheese}) = \frac{36}{120} = \frac{3}{10}$ | ✓ $\frac{x}{120}$<br>✓ $\frac{120-x}{119}$<br>✓ $\frac{x}{120} \times \frac{120-x}{119} = \frac{18}{85}$<br><br>✓ $x = 84 \quad \text{or} \quad x = 36$<br><br>✓ $\frac{3}{10}$<br>(5) |
|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | [13]                                                                                                                                                                                   |

**TOTAL/TOTAAL: 150**